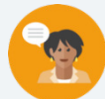


11.5 Solving Real-World Problems With Rational Numbers – Part 1

Lesson Introduction

 Benchmark	MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.		 Learning Target	I can solve problems with one or two operations by adding, subtracting, multiplying, and/or dividing rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
 Essential Questions	- How do you solve multi-step problems involving addition and subtraction of rational numbers? - How do you solve multi-step real-world problems using multiplication and division of rational numbers?			
 Lesson Narrative	Lesson 5 opens a series of lessons on solving problems using any of the four operations in a variety of real-world contexts. Students focus on problems involving just one or two operations before extending to more complex scenarios in Lessons 6 and 7.			
 Component Summary	11.5.1 Warm-Up (~5 min.)	Identify patterns in division equations (<i>SE p. 185</i>)	11.5.3 Guided Instruction (10-15 min.)	Using key information and drawing diagrams, break into steps to solve real-world, multi-step problems with rational numbers (<i>SE p. 186</i>)
	11.5.2 Collaboration (10-15 min.)	Draw a diagram to solve real-world problems with rational numbers (<i>SE p. 185</i>)	11.5.4 Your Turn! (10-15 min.)	Solve real-world, multi-step problems with rational numbers (<i>SE p. 188</i>)
	11.5.5 Wrap-Up (~5 min.)	Sometimes, always, or never growth mindset reflection (<i>SE p. 189</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Credit - Debit - Equations - Expressions		(Optional) Chart paper (Optional) Markers - Calculator	N/A

 Teacher Tips	Common Misconceptions • Students may have trouble representing values in the questions with expressions and equations. • Students may struggle to add and subtract mixed numbers with different denominators. • Students may not recognize that when asked to describe how a value changes after a decrease, a negative number should be used.	Things to Consider • Consider your students' experience with banks and checking accounts. If needed, discuss what students know about how bank accounts work, and have them read through the question to note any language they are unfamiliar with. • Allowing students to use calculators during the lesson will help students focus on the real-world contexts. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads 11.5.2 Collaboration presents students with a lot of text to work through when solving the given question. Consider utilizing “Three Reads” when presenting the question, positioning students to make sense of the context, highlight numbers and quantities, and identify key information in the question before beginning to solve.	MA.K12.MTR.7.1: Real-World Contexts Students engage in real-world problem solving throughout the lesson. MTR 7 asks students to “Use models and methods to understand, represent and solve problems.” Students utilize ‘Three Reads’ in this lesson to help them make sense of the question as they read for context, to highlight numbers and quantities, and to identify key information, drawing diagrams. and breaking the questions down into simpler steps to solve.
 Differentiate Instruction	Students can engage with On-Ramp videos on the following topics for additional support: • Real-World Multi-Step Addition and Subtraction Problems • Subtract Fractions with Unlike Denominators (11.5.3 Guided Instruction question 2)	
Lesson Notes:		

11.5.1 Warm-Up: Notice and Wonder

Component Overview: Students engage in a Notice and Wonder routine to examine patterns in division equations.



11.5.1 Warm-Up: Notice and Wonder

Several division equations are shown.

$$0.28 \div 0.07 = \square$$

$$2.8 \div 0.7 = \square$$

$$28 \div 7 = 4$$

$$280 \div 70 = 4$$

$$2,800 \div 700 = 4$$

$$28,000 \div 7,000 = 4$$

1. What do you wonder about the pattern?

Answers will vary. Will the pattern continue with the first two expressions? How does the decimal point affect the solution?

2. What do you notice about the equations?

Answers will vary. The solution to all the expressions in the pattern is 4.

3. Use the pattern in the equations to determine the quotients of the first two equations.

$$0.28 \div 0.07 = 4$$

$$2.8 \div 0.7 = 4$$



Notice and Wonder

Notice and Wonder activities allow students to lead the conversation. Encourage students to notice that the decimal place is changing in each equation while the quotient remains the same. Highlight student responses that recognize this and wonder what that means for the missing quotients.

11.5.2 Collaboration: Solving a Problem with Rational Numbers

Component Overview: Students work together to solve a real-world, multi-step problem with rational numbers. Students draw a diagram to model the problem and use the diagram to help answer questions about the problem.

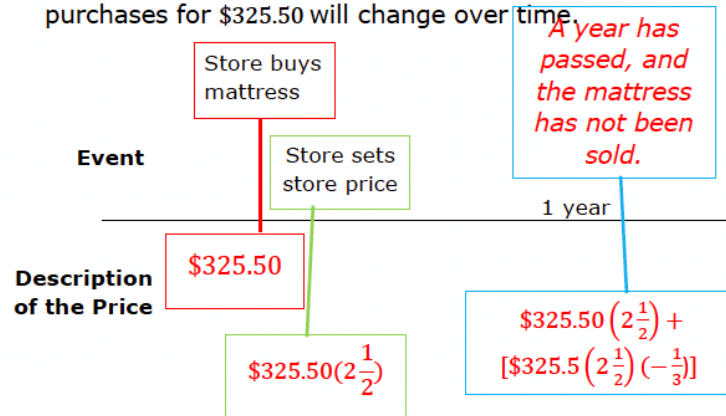


11.5.2 Collaboration: Solving a Problem with Rational Numbers

In order to make a profit, stores sell products for more than what they paid for them. For example, a local furniture store buys a mattress from a manufacturer for a certain price. The store then sets the store price at $2\frac{1}{2}$ times what they paid the manufacturer for it. If the mattress hasn't sold in a year, then the store will reduce its price by $\frac{1}{3}$ of the original store price.

Work with your partner to complete the following based on the information given.

1. Drawing a diagram can help make sense of the problem. Without calculating, complete the timeline to show how the price of a mattress the store purchases for \$325.50 will change over time.



Three Reads

Consider engaging students in 'Three Reads' to prepare them to make sense of the question.

Begin by reading the question aloud to the whole group after posing the prompt, "What is this situation about?" Students sit with their partner and discuss, sharing with the whole group when finished.

Pose the second prompt, "What are the quantities in this situation?" Have partners read through the second time and bring the whole group together again; list quantities identified on chart paper or whiteboard.

Pose the third prompt, "What mathematical questions can we ask about this situation?" Have partners again read through for a third time, asking them to come up with two questions. Once again, bring the whole group together to share their questions. Consider prompting students to think about the context and identified quantities by asking, "Can this question be answered using the information we've already identified in the problem?"

At this point, partners begin to collaborate to work through and answer the questions.

11.5.2 Collaboration: Solving a Problem with Rational Numbers (continued)

2. Explain whether the store's first price will be greater than or less than the price they paid the manufacturer.

Answers will vary. The store's first price will be greater than the price the local furniture store paid the manufacturer because the store marked up the price $2\frac{1}{2}$ times.

3. What is the first price the store will set for the mattress?

\$813.75

4. Explain whether the price after a year will be greater than or less than the store's first price.

Answers will vary. The price after a year will be less than the store's first price because the store reduces the price by $\frac{1}{3}$ of the original store price to increase the chance of selling the mattress.

5. What is the sale price of the mattress after one year?

\$542.50

6. If the store sells the mattress after one year, explain if the store will make or lose money on the mattress.

Answers will vary. The store will make a profit from the original purchase of the mattress for \$325.50 after the original increase and reduction in price after a year.



Refer to information recorded in 'Three Reads' routine and the timeline students filled in for question 1:

- What quantity describes how the store sets the price for the mattress?
- Which price represents what the store paid for the mattress?
- How does the price change once the store sets the price for the mattress?
- Which quantity represents the change in the value of the mattress after one year?



Encourage students to answer one or two of the identified questions in the 'Three Reads' practice at the beginning of 11.5.2 Collaboration.

11.5.3 Guided Instruction: Solving Real-World Problems with Rational Numbers

Component Overview: Students solve real-world, multi-step problems with rational numbers by pulling out key information, drawing diagrams, and breaking it down into each step.



11.5.3 Guided Instruction: Solving Real-World Problems with Rational Numbers

1. Alexis has a monthly subscription to the video streaming service ViewNow for \$7.20 per month. For her birthday she received a ViewNow gift card in the amount of \$55.00.
 - a. Write an expression using addition that represents the balance on the card after one month if the only charge on the gift card is the monthly subscription.
 $55 + (-7.20)$
 - b. Complete the table by filling in values on each blank line to record the activity on the account each month. Continue each month to determine how many months it will take before Alexis does not have enough money to pay the monthly fee using the gift card. Part of Month 1 has been completed for you.

Month	Starting Balance	Change in Account	Ending Balance
1	\$55.00	$\$55.00 + (-\$7.20)$	\$47.80
2	\$47.80	$\$47.80 + (-\$7.20)$	\$40.60
3	\$40.60	$\$40.60 + (-\$7.20)$	\$33.40
4	\$33.40	$\$33.40 + (-\$7.20)$	\$26.20
5	\$26.20	$\$26.20 + (-\$7.20)$	\$19.00
6	\$19	$\$19.00 + (-\$7.20)$	\$11.80
7	\$11.80	$\$11.80 + (-\$7.20)$	\$4.60
8			
9			

- c. How many months will it take before Alexis does not have enough money left on the gift card to pay the monthly charge?
7 Months



Students may decide the service fee is subtracting the positive number (the fee), while other students may decide to add the fee as a negative value. Both ways work, but make sure they understand not to both subtract and use a negative number to represent the fee. The number of rows given on the page is not an indication of the answer. There may be empty rows.



The tables provide a visual means for students to organize the information presented in the question, providing a visual entry point for learners. Encourage students to discuss their strategies for solving with a partner to provide additional auditory entry points.

11.5.3 Guided Instruction: Solving Real-World Problems with Rational Numbers (continued)

2. The **departure from the average** is the difference between the actual amount of rain and the average amount of rain for a given month. The departure from the average rainfall, in inches (in.), for Orlando, Florida for June, July, and August is shown in the table.

June	July	August
$-3\frac{19}{20}$ in.	$-1\frac{7}{10}$ in.	$4\frac{5}{8}$ in.

- a. If the average rainfall for June is $7\frac{29}{50}$ in., what was the actual rainfall in June?
 $3\frac{63}{100}$
- b. What is the difference in the departure from the average rainfall from July to August?
 $4\frac{5}{8} - (-1\frac{7}{10}) = 6\frac{13}{40}$
3. Ramon paid \$18,360.00 for a new car. By the end of the first year, the value of the car changed by $-\frac{3}{14}$ of its original value. By the end of the second year, the value of the car changed by $-\frac{2}{9}$ of its value at the end of the first year.
- a. Is the value of the car increasing or decreasing during these two years?
The value of the car is decreasing during the two years after purchase.
- b. By what amount did the value of the car change at the end of the first year?
 $-\$3,934.29$
- c. What is the value of Ramon's car at the end of the second year?
 $\$11,220$
4. A water well drilling rig dug to a depth of -147.5 feet (ft). The rig drills $-2\frac{1}{2}$ ft in one hour.
- a. How many hours has the drill been running?
 $-147.5 \div (-2.5) = 59 \text{ hours}$
- b. At this rate, how many hours will it take until the drill reaches a depth of -250 ft?
 $-250 \div (-2.5) = 100 \text{ hours}$



Departure from the average may be a new term for students but they have likely used the idea before when comparing values to averages. Pause to have students consider contexts where they may have used the idea themselves. One possible context is grades. If a student makes an 85 on a test but their test average is 91 then the 85 is a difference of 6 from the average.

Students may struggle to find the departure from the average. Ask guiding questions to help students subtract mixed numbers with different denominators:

- What is the least common multiple for these denominators? How can you use LCM to find the departure from the average?
- What operation can you use to find the departure from the average? Why does that work?

Provide students with some background information on depreciation. Most goods have a depreciation. Some goods may depreciate and then after time their value increases because they become antiques. Explain to students that the $-\frac{3}{14}$ means that this is the amount of the value that the car loses during its first year. Another way of stating that is that at the end of the first year the car is $\frac{11}{14}$ of its original value.

Students may not identify the value of the car after the first year decreased, so the change in value is represented by a negative number. Consider asking which students wrote a positive value and which wrote a negative value. Then, choose one student from each to explain their reasoning. If students state that the car is depreciating \$3,934.29 then the negative is inferred.

11.5.4 Your Turn!

Component Overview: Students solve real-world, multi-step problems with rational numbers.



11.5.4 Your Turn!

1. The table shows the change in value for a share of stock in a company at the end of each week during the month of December. A negative value represents the amount the share decreased, and a positive value represents the amount the share increased.

December
-\$22.65
\$225.30
-\$106.42
\$91.10

If the value of the stock began at \$75.37 at the start of the month, what is the final value of the share at the end of the month?

$$\$75.37 - \$22.65 + \$225.30 - \$106.42 + \$91.10 = \$262.70$$

2. Samuel is practicing for his town's annual high school quarterback football-throwing competition. Last year's champion threw the football $40\frac{1}{4}$ yards (yd). A summary of Samuel's throws in relation to the throws of last year's champion is shown. Complete the table to determine how far Samuel threw the football during each of his tries.

Throw	Comparison of Samuel's Throw to Last Year's Champion	Expression	Yards Thrown
1 st	$+3\frac{1}{3}$ yards	$40\frac{1}{4} + 3\frac{1}{3}$	$43\frac{7}{12}$
2 nd	$-\frac{9}{7}$ yards	$40\frac{1}{4} + \left(-\frac{9}{7}\right)$	$38\frac{27}{28}$
3 rd	$-2\frac{6}{13}$ yards	$40\frac{1}{4} + \left(-2\frac{6}{13}\right)$	$37\frac{41}{52}$



The contexts may not be familiar to students. Encourage students to use the information in the context to determine what to do. Placing students in new situations helps them to apply the content they have learned. Allow students to have productive struggle. Direct students with questions that may help them retrieve the information themselves.

As you circulate note whether students are struggling with operations or with the context. Use this information to inform your instruction on the remaining lessons of this section.

11.5.4 Your Turn! (continued)

3. Delaney wants to sell her bicycle. The original price of the bike was \$150. The value of the bike has changed by $-\frac{2}{5}$ of the original price. What is the current value of the bike?

\$90

4. At the beginning of the day on Monday, Stephon had \$15.78 in his lunch account. At the end of the day on Friday, he had \$1.53 in his lunch account. If Stephon purchased the same lunch each day on Monday through Friday, describe how the account changed each day.

The account decreased by \$2.85 per day.



Encourage students who may be struggling with question 4 to refer to the diagram they used in 11.5.2 Collaboration question 1.

11.5.5 Wrap-Up: Growth Mindset

Component Overview: Students will use the terms “sometimes,” “always,” and “never” to describe how each statement represents their feelings in relation to a growth mindset.



11.5.5 Wrap-Up: Growth Mindset

- As you reflect on today’s lesson, circle the word *Sometimes*, *Always*, or *Never* to indicate how often each statement describes you.

Answers will vary.

When I work hard on a problem, I expect that the answer will be correct.	Sometimes	Always	Never
There is no point in trying to do a problem if I know I won’t get the correct answer.	Sometimes	Always	Never
Even if I made a mistake, I can still learn from working through a problem.	Sometimes	Always	Never
I can show that I understand how to solve a problem, even if there is an error in my work.	Sometimes	Always	Never
If my answers are incorrect even after working extremely hard, I still choose to work hard on the next problem.	Sometimes	Always	Never



Consider prompting students to choose one of the statements and provide an example and justification for their response. Encourage them to use an example from this lesson or Unit 11.

Use student feedback to help you consider which students may need to be encouraged to embrace a growth mindset.